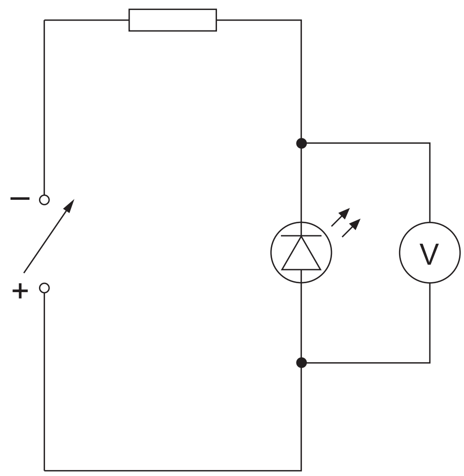
7. (a) Lewis uses the following circuit to find the minimum pd, V_{min}, across an LED at which light is emitted by the diode. He collects his data in a dark room where he varies the pd of the power supply until the LED lights.



Lewis measures the minimum pd, V_{min}, five times and his results are shown below.

V_{min} / V	1.81	1.87	1.93	1.84	1.90

(i) Determine the mean value for V_{min} along with its **percentage** uncertainty. [3]

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(ii) Lewis uses the following equation to calculate the wavelength of the light expected to be produced by the LED:

$$eV_{\text{min}} = \frac{hc}{\lambda}$$

Calculate the mean value for λ and its **absolute** uncertainty. [3]

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(b) Lewis noted the value for V_{min} when he noticed that the LED had turned on. Suggest an improvement to his method. [1]

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